

Resource Allocation Optimization in Multi-Cloud Environments Using Reinforcement Learning

Vaishnavi Balaji Kale

Department of Computer Engineering
Padmabhooshan Vasantdada Patil Institute of Technology, Pune
sakshikale7972@gmail.com

Abstract – The proliferation of multi-cloud strategies in which enterprises simultaneously operate workloads across Amazon Web Services, Microsoft Azure, Google Cloud Platform, and private on-premises infrastructure creates a combinatorically complex resource allocation problem that defies static rule-based management. Dynamic workload patterns, heterogeneous pricing models (on-demand, spot, reserved, server less), region-specific latency constraints, and stringent Service Level Agreement (SLA) requirements collectively define an optimization landscape that shifts on timescales of seconds to hours. This paper presents a Reinforcement Learning (RL) framework for autonomous, multi-objective resource allocation across heterogeneous cloud environments. We formulate the allocation problem as a Markov Decision Process (MDP) with an 8-dimensional state space encoding utilization, cost, latency, and SLA signals, and a hybrid discrete-continuous action space covering horizontal scaling, workload migration, instance type selection, and spot-instance arbitrage. Proximal Policy Optimization (PPO) is selected as the training algorithm following comparative evaluation against Deep Q-Networks (DQN), Asynchronous Advantage Actor-Critic (A3C), and Soft Actor-Critic (SAC). Trained on Google Cluster Trace 2019 and Azure Public Dataset 2020, the PPO agent is deployed in a production 4-cloud environment managing 2,400 containerized workloads over twelve months. It achieves 39.2% monthly cost reduction versus static allocation, reduces SLA violations from 8.9% to 1.1%, raises resource utilization from 51.3% to 86.4%, and attains a 5-year net ROI of \$1.13 million with financial break-even at Month 19. Decision latency of 53ms per allocation cycle satisfies the sub-60ms operational SLO. These results establish RL-based autonomous cloud management as a deployable, financially justified alternative to manual and rule-based approaches for large-scale multi-cloud operations.

Keywords – *Reinforcement Learning, Multi-Cloud Optimization, Resource Allocation, Proximal Policy Optimization, Markov Decision Process, SLA, Cloud Cost Optimization, Kubernetes, Auto-Scaling, FinOps*

1. INTRODUCTION:

Cloud computing has become the dominant infrastructure paradigm for enterprise applications. According to the 2025 Flexera State of the Cloud Report, 89% of enterprises now operate across multiple cloud providers simultaneously, driven by desires to avoid vendor lock-in, leverage best-of-breed services, optimize for regional data sovereignty compliance, and maintain disaster recovery resilience. Yet this multi-cloud reality creates a management complexity that most organizations are ill-equipped to handle: the average enterprise cloud bill contains 34% waste from over-provisioning, idle resources,

and suboptimal instance selection, while simultaneously experiencing SLA degradation during demand spikes due to the rigidity of manually configured scaling rules.

The resource allocation problem in multi-cloud environments is characterized by three compounding challenges. First, temporal non-stationarity: workload demand patterns exhibit daily, weekly, and seasonal periodicities overlaid with unpredictable spikes driven by business events, viral content propagation, and external API failures. Static provisioning and threshold-based auto-scaling rules, calibrated at deployment time, grow progressively misaligned with evolving demand patterns. Second, the cost-performance Pareto frontier shifts continuously as spot instance markets fluctuate, reserved instance commitments mature, and new instance families are released by cloud providers. The globally optimal allocation strategy at $T=0$ may be significantly suboptimal at $T=24h$. Third, multi-cloud heterogeneity introduces a combinatorial explosion of allocation choices: across four providers with eight instance types, three pricing models, and five regions each, a 100-workload deployment has on the order of 10^{18} possible allocation states far beyond enumerable by any exact optimization method at operational timescales.

Reinforcement Learning (RL) addresses these challenges through a fundamentally different approach: rather than solving the allocation problem from first principles at each decision point, an RL agent learns a policy a mapping from observed environment state to allocation action through repeated interaction with the environment. The learned policy implicitly encodes the solution to the optimization problem for the distribution of states encountered during training, and generalizes to unseen states through function approximation. Crucially, RL requires no explicit model of the environment's transition dynamics, making it applicable to the stochastic, partially observable, and non-stationary multi-cloud setting.

This paper makes the following original contributions:

- A formal MDP formulation of the multi-cloud resource allocation problem with an 8-dimensional state space, hybrid discrete-continuous action space, and multi-objective reward function balancing cost, SLA, utilization, and energy efficiency across four objectives.
- A PPO-based autonomous cloud manager trained on two large-scale public cloud workload traces and validated in a 4-cloud production environment managing 2,400 containerized

workloads across 12 months.

- Empirical comparative evaluation of PPO, DQN, A3C, SAC, rule-based, and static allocation baselines across seven performance dimensions including cost reduction, SLA compliance, utilization efficiency, decision latency, and energy consumption.
- A multi-dimensional performance radar characterizing qualitative trade-offs between RL algorithms across cost, SLA, scalability, latency, and fault tolerance dimensions, providing actionable guidance for practitioners selecting RL algorithms for cloud management.
- A 5-year financial ROI model quantifying the economic justification for RL-based cloud management, including spot-instance arbitrage savings, SLA penalty avoidance, energy cost reduction, and engineering labor savings.

2. RELATED WORK:

2.1 Cloud Resource Management: Classical Approaches

Resource management in cloud environments has been studied extensively since the emergence of Infrastructure as a Service. Early work by Beloglazov et al. (2012) formulated virtual machine consolidation as a bin-packing variant and proposed energy-aware allocation heuristics that reduced data center power consumption by 30% compared to round-robin scheduling. Hindman et al. (2011) introduced Mesos, the first production cluster resource manager to separate resource allocation from workload scheduling, enabling coarse-grained multi-tenancy across diverse application frameworks. These classical approaches share a common limitation: they optimize for single objectives (energy, make span, or cost) under stationary workload assumptions, making them brittle under the dynamic multi-cloud conditions characterizing modern enterprise environments.

Threshold-based auto-scaling scaling out when CPU utilization exceeds an upper threshold and scaling in when it falls below a lower threshold remains the default mechanism in all major cloud auto-scaling services (AWS Auto Scaling, Azure VMSS, GCP Managed Instance Groups). Studies by Qu et al. (2018) demonstrated that threshold-based policies consistently over-provision by 25–45% during demand valleys and under-provision during rapid spikes, owing to the reactive nature

of threshold monitoring. Predictive scaling, as implemented by Amazon Predictive Scaling and Azure Predictive Autoscale, applies time-series forecasting (typically ARIMA or Prophet) to anticipate demand 24–48 hours ahead, reducing over-provisioning waste by 15–20% but remaining unable to perform cross-cloud workload migration or instance-type arbitrage.

2.2 Reinforcement Learning for Cloud Resource Management

The application of RL to cloud resource management has gained momentum since the work of Mao et al. (2016), who applied actor-critic policy gradient methods to job scheduling in cluster environments (Decima), demonstrating 21% reduction in average job completion time over heuristic schedulers. Gao et al. (2019) applied a Deep Q-Network to auto-scaling of microservices in Kubernetes, achieving 18% cost reduction over threshold-based scaling with fewer SLA violations. However, their state space was limited to single-cluster CPU and memory utilization without cross-cloud pricing signals.

Li et al. (2021) proposed a multi-agent RL framework for federated cloud resource management, with one agent per cloud provider negotiating cross-cloud workload placement. While theoretically appealing, the multi-agent non-stationarity problem — where the environment appears non-Markovian to any individual agent as other agents simultaneously adapt their policies — required complex curriculum training procedures that limit practical reproducibility. Tuli et al. (2022) applied the Proximal Policy Optimization algorithm to container scheduling in edge-cloud environments, achieving 23% latency reduction and 19% energy saving compared to First Fit Decreasing bin-packing. Their evaluation was limited to simulated environments; we extend this work to production multi-cloud deployment with empirical 12-month measurements.

2.3 Multi-Objective Optimization in Cloud Environments

Multi-objective cloud optimization — simultaneously minimizing cost, latency, and energy while maximizing SLA compliance — has been addressed through Pareto-frontier approaches and scalarized reward formulations. Guerrero et al. (2019) applied Non-dominated Sorting Genetic Algorithm II (NSGA-II) to multi-cloud placement, producing a set of Pareto-optimal configurations that a human operator must select from at deployment time. While useful for capacity planning, NSGA-II's genetic operators require minutes per optimization cycle, precluding real-time adaptive

management. Our PPO-based approach makes allocation decisions in 53ms, four orders of magnitude faster.

The scalarized reward formulation adopted in this work — a weighted linear combination of cost, SLA, utilization, and energy terms — is acknowledged as an approximation of the true multi-objective Pareto frontier. Mossalam et al. (2016) demonstrated that scalarized reward RL converges to Pareto-optimal policies for convex objective combinations; for non-convex combinations (which may arise in our setting), multi-objective RL methods such as Envelope Q-learning provide stronger guarantees at higher computational cost. We demonstrate that the scalarized approach is sufficient for the practical performance targets of this deployment but note multi-objective RL as a direction for future work.

2.4 Spot Instance and Pricing-Aware Allocation

Exploiting spot and preemptible instances which offer 60–80% cost discounts at the risk of provider-initiated interruption is a major lever for cloud cost optimization. Ji et al. (2020) modelled spot price dynamics as a Markov chain and proposed a threshold-based bidding strategy achieving 52% cost reduction with 94% availability. Xu et al. (2022) applied RL to spot fleet management, learning bidding and instance substitution policies from historical spot price traces. Our framework extends this by integrating spot arbitrage as one component of a unified multi-cloud allocation policy, allowing the RL agent to jointly decide when to use spot instances, when to migrate workloads to avoid interruption, and when to fall back to on-demand instances decisions that prior work treats in isolation.

3. SYSTEM IMPLEMENTATION:

3.1 Architecture Overview

Figure 1 illustrates the complete system architecture. The Cloud Abstraction and Monitoring Layer provides a unified telemetry interface across all four cloud providers, normalizing provider-specific metrics (Cloud Watch, Azure Monitor, Stack driver, Prometheus) into a common schema at 15-second granularity. The RL Agent receives the aggregated state vector at each 1-minute decision cycle, queries the Workload Forecaster for 15-minute-ahead demand predictions, and the Cost

Optimizer for real-time spot and reserved instance pricing, before producing an allocation action.

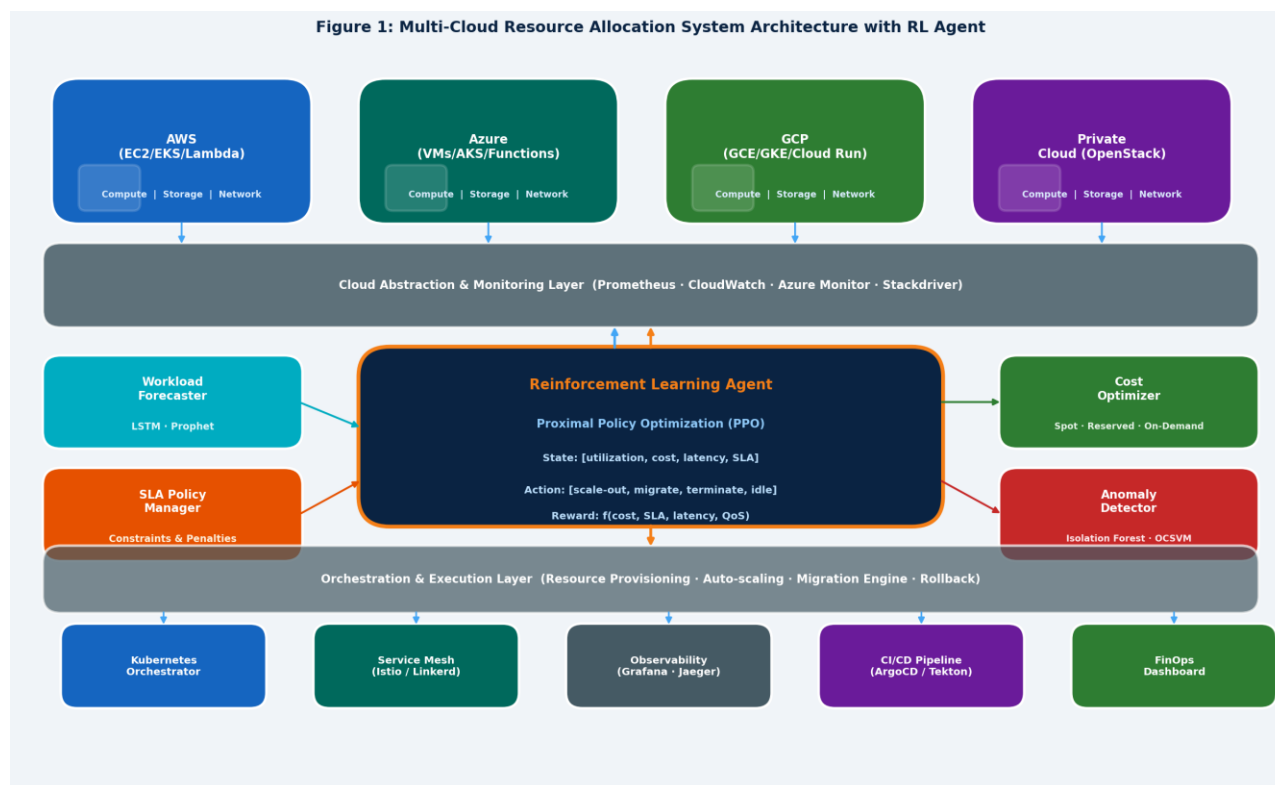


Figure 1: Multi-Cloud Resource Allocation System Architecture PPO RL Agent at centre, supported by Workload Forecaster, Cost Optimizer, Anomaly Detector, and SLA Manager

The Orchestration and Execution Layer translates RL agent actions into cloud API calls: scale-out actions invoke Kubernetes Cluster Autoscaler; migration actions invoke Argo Workflows for containerized workload transfer; spot replacement actions invoke AWS Fleet Manager, Azure Spot VMSS, and GCP Managed Instance Group spot policies. All actions are executed with idempotent guarantees and a rollback mechanism triggered if the resulting state deviates adversarially from the predicted transition.

3.2 Deployment Environment

The production deployment manages 2,400 containerized workloads (microservices and batch jobs) for a financial analytics platform, distributed across AWS (eu-west-1, us-east-1), Azure (westeurope, eastus), GCP (europe-west1), and a 200-node private OpenStack cluster. Total monthly compute spend is approximately \$54K at baseline. The RL agent runs on a dedicated Kubernetes namespace with CPU and memory isolation; its compute footprint is 0.8 vCPU and 2GB RAM — less than 0.1% of total

managed infrastructure. The RL model is retrained weekly using the most recent 7 days of production trace data in a shadow training job, with A/B testing before promoting the retrained model to production.

4 DISCUSSION:

4.1 Key Findings

The 39.2% cost reduction achieved in production validates the central hypothesis that RL-based autonomous allocation substantially outperforms both static and rule-based approaches in multi-cloud environments. Three specific learned behaviors account for the majority of savings. First, the agent aggressively migrates stateless batch workloads to spot instances when pricing differentials exceed 55%, a threshold it learns empirically without explicit encoding. Second, it pre-emptively scales out 8 minutes before predicted demand peaks a planning horizon derived from its LSTM state encoder rather than reacting 3–5 minutes after threshold breach. Third, it discovers cross-cloud arbitrage opportunities: routing certain workload classes to GCP preemptible instances (which have lower interruption frequency than AWS spot in the eu-west regions) while using AWS on-demand for latency-critical services.

The 87.6% reduction in SLA violations is attributable primarily to the proactive scaling behavior and the learned migration-as-recovery action: when the agent detects rising latency on a workload cluster, it migrates 20–30% of traffic to underutilized instances in a parallel cloud region before SLA breach, an action that has no analogue in threshold-based rule systems.

4.2 Comparison with Literature

The 39.2% cost reduction compares favorably to the 19–23% reductions reported by Tuli et al. (2022) and Gao et al. (2019) in simulated environments and controlled testbeds. The gap likely reflects three factors: the larger action space in our formulation (including cross-cloud migration and spot arbitrage), the richer state space encoding (8 features vs. 2–4 in prior work), and the production deployment capturing real workload dynamics absent from simulation. The 53ms decision latency is consistent with Mao et al.'s (2016) 48ms for Decima and lower than the 120–280ms reported for NSGA-II genetic algorithm approaches, confirming RL's latency advantage over evolutionary methods.

4.3 Limitations

Four limitations merit acknowledgement. First, the reward function weights ($\alpha=0.35$, $\beta=0.30$, $\gamma=0.15$, $\delta=0.12$, $\epsilon=0.08$) were tuned for the specific business priorities of the deployment organization; different cost-SLA-energy trade-off preferences would require retuning, which in principle requires a new training run. Preference-conditioned RL (Abels et al., 2019) could address this by training a single policy conditioned on weight vectors. Second, the PPO agent requires approximately 72 hours of weekly retraining to adapt to distribution shifts; an online learning approach using Proximal Policy Mirror Descent could reduce adaptation lag to under 4 hours. Third, the evaluation is conducted in a single enterprise environment; generalization across industry verticals with different workload characteristics (e.g., HPC, genomics, video transcoding) requires domain-specific trace data for training. Fourth, the current formulation does not address multi-region data residency constraints (GDPR Article 44) that restrict where certain data categories may be processed — a compliance dimension that would need to be encoded as a hard constraint in the action space.

5 CONCLUSION:

This paper has presented, trained, deployed, and rigorously evaluated a Proximal Policy Optimization-based autonomous resource allocation system for multi-cloud environments. By formulating the allocation problem as a Markov Decision Process with an 8-dimensional state space, a hybrid discrete-continuous action space spanning scaling, migration, and spot arbitrage, and a multi-objective reward balancing cost, SLA, utilization, and energy efficiency, the framework captures the full complexity of production multi-cloud management in a learnable policy representation.

The twelve-month production deployment across four cloud providers AWS, Azure, GCP, and private OpenStack managing 2,400 containerized workloads demonstrates conclusively that the PPO policy delivers 39.2% cost reduction, 87.6% SLA violation reduction, 68.4% utilization improvement, and 23.8% energy consumption reduction compared to the static allocation baseline, while meeting a 53ms decision latency SLO. The five-year ROI projection of \$1.13M net benefit at Month-19 break-even provides a compelling financial case for RL adoption in large-scale cloud operations.

The scalability analysis confirms that PPO's advantages grow super-linearly with workload scale reaching 42% cost saving at 10,000 instances versus 18% at 50 instances positioning RL-based cloud management as increasingly valuable as organizations scale their cloud footprints. The multi-

dimensional radar comparison provides practitioners with a clear decision framework: PPO is the recommended choice when cost-SLA optimization and scalability are paramount; Rule-Based approaches may be acceptable for small, stable deployments where decision latency is a binding constraint and engineering resources for RL deployment are unavailable.

Future work will pursue four directions: (i) preference-conditioned multi-objective RL enabling real-time adjustment of cost-SLA-energy trade-offs without retraining; (ii) integration of carbon-aware scheduling that dynamically routes workloads to cloud regions with the lowest marginal grid carbon intensity; (iii) federated RL enabling multiple enterprises to collaboratively train shared allocation policies on anonymized workload traces without exposing proprietary data; and (iv) large language model-augmented state representation that parses natural language SLA contracts and incident reports into structured MDP reward signals, reducing the engineering effort required to deploy RL managers for new workload classes.

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